

Next-generation Bayesian local robustness

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Copy over MrP analysis

- Meaningful Bayesian local specification diagnostics
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
- Flow-based Bayesian local robustness

Negative binomial experiment

Example: Negative binomial models with an unknown parameter γ .

For $n = 1, \dots, N$ let $x_n | \gamma \stackrel{iid}{\sim} \text{NegativeBinomial}(r, \gamma)$ for fixed r .

$$\text{Write } \log p(X | \gamma, w) = \sum_{n=1}^N w_n \ell_n(\gamma).$$

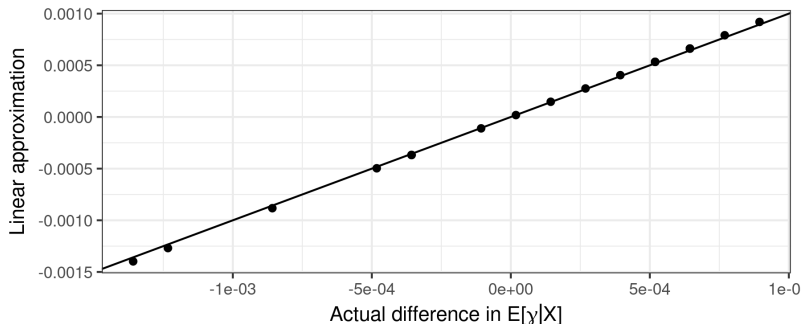
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Negative Binomial model
leaving out single datapoints with $N = 800$



Variance consistency theorem

How do the results for a single weight translate into variance estimates?

$$\text{Var}_{p(w)} \left(\mathbb{E}_{p(\theta|X,w)} [f(\theta)] \right) = \frac{1}{N^2} \sum_{n=1}^N \left(\psi_n - \bar{\psi} \right)^2 + \text{Term involving } \mathcal{E}_n(w) \text{ for } n = 1, \dots, N$$

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- Assume (sketch): A well-behaved MAP *maximum a posteriori* estimator $\hat{\theta}$ exists.
 - The dimension of θ is fixed as $N \rightarrow \infty$
 - The expected log likelihood has a strict maximum at θ_∞
 - The observed log likelihood satisfies $\hat{\theta} \rightarrow \theta_\infty$
 - The expected log likelihood Hessian is negative definite at θ_∞
- Assume (sketch): We can apply standard asymptotics.
 - The log prior and log likelihood are four times continuously differentiable
 - The prior is proper, and a technical set of prior expectations are finite
 - The log likelihood and its derivatives are dominated by a square-integrable envelope function for all θ in a neighborhood of θ_∞ .

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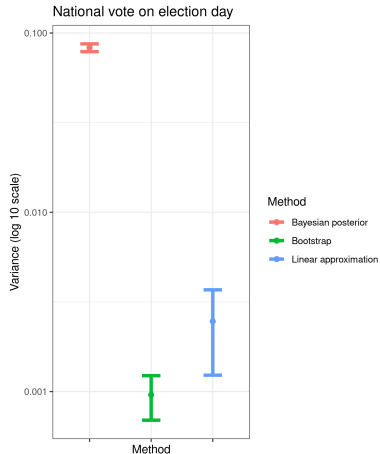
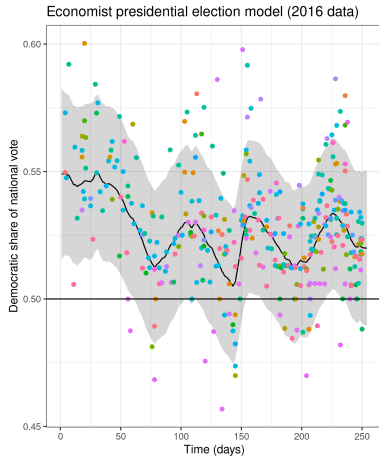
Theorem 2 [Giordano and Broderick, 2023]: Under the above assumptions,

$$\sqrt{N} \left(\mathbb{E}_{p(\theta|X)} [g(\theta)] - g(\theta_\infty) \right) \xrightarrow[N \rightarrow \infty]{dist} \mathcal{N}(0, V^g) \quad [\text{Kleijn and Van der Vaart, 2012}]$$

$$\text{and } V^{\text{IJ}} := \frac{1}{N} \sum_{n=1}^N \left(\psi_n - \bar{\psi} \right)^2 \xrightarrow[N \rightarrow \infty]{prob} V^g. \quad (\text{Our contribution})$$

How to connect to the election data?

Problem: MCMC is only interesting when the posterior doesn't concentrate.



Example: Exponential families with random effects (REs) λ and fixed effects γ .

If the observations per random effect remains bounded as $N \rightarrow \infty$, then

- Parameter λ (“local”) grows in dimension with N .
- Parameter γ (“global”) is finite-dimensional.
- Marginally $p(\lambda|X)$ does not concentrate.
- Marginally, $p(\gamma|X)$ concentrates.

High dimensional problems

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- Marginally, $p(\gamma|X)$ concentrates.

In general, we cannot hope for an asymptotic analysis of $\mathbb{E}_{p(\lambda, \gamma|X)} [f(\lambda)]$.

Can we save the approximation when *some* parameters concentrate?

Does the residual vanish asymptotically for $w_n \mapsto \mathbb{E}_{p(\gamma|X; w_n)} [f(\gamma)]$?

High dimensional problems

We assume that $p(\gamma|X)$ concentrates but $p(\lambda|X)$ does not. By our series expansion:

$$\begin{aligned} \mathbb{E}_{p(\gamma, \lambda|X; w_n)}[\gamma] - \mathbb{E}_{p(\gamma, \lambda|X)}[\gamma] = \\ \psi_n(w_n - 1) + \mathcal{E}_n(w) \end{aligned}$$

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Corollary [Giordano and Broderick, 2023]:

In general, $w_n \mapsto N \left(\mathbb{E}_{p(\gamma|X; w_n)} [\gamma] - \mathbb{E}_{p(\gamma|X)} [\gamma] \right)$ remains non-linear as $N \rightarrow \infty$.

Example: Poisson regression with Gamma-distributed random effects

For $g = 1, \dots, G$, $\lambda_g \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$ for fixed α, β

For $n = 1, \dots, N$, $g_n \stackrel{iid}{\sim} \text{Categorical}(1, \dots, G)$, $y_n | \lambda_n, \gamma, g_n \stackrel{iid}{\sim} \text{Poisson}(\gamma \lambda_{g_n})$.

$x_n = (y_n, g_n)$ are IID given λ, γ . Write $\log p(X | \lambda, \gamma; w) = \sum_{n=1}^N w_n \ell_n(\lambda, \gamma)$.

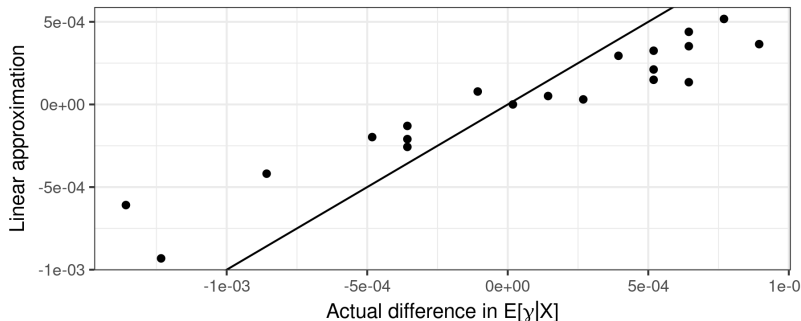
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Poisson random effect model
leaving out single datapoints with $N = 800$



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Bayesian von-Mises Expansion

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Define the “generalized posterior” functional (θ may be growing in dimension):

$$T(\mathbb{G}, N) := \frac{\int g(\theta) \exp \left(N \int \ell(x_0|\theta) \mathbb{G}(dx_0) \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(x_0|\theta) \mathbb{G}(dx_0) \right) \pi(\theta) d\theta}.$$

Let $\mathbb{F}_N$ denote the empirical distribution over x_n . Then

$$\mathbb{E}_{p(\theta|X)}[g(\theta)] = \frac{\int g(\theta) \exp \left(N \frac{1}{N} \sum_{n=1}^N \ell(x_n|\theta) \right) \pi(\theta) d\theta}{\int \exp \left(N \frac{1}{N} \sum_{n=1}^N \ell(x_n|\theta) \right) \pi(\theta) d\theta} = T(\mathbb{F}_N, N).$$

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Let $\mathbb{F}$ denote the true distribution of x_n , and let $\mathbb{F}_N^t = t\mathbb{F}_N + (1-t)\mathbb{F}$.

We can study the *von Mises expansion*:

$$\begin{aligned} \sqrt{N} \left(\mathbb{E}_{p(\theta|X)}[g(\theta)] - T(\mathbb{F}, N) \right) &= \sqrt{N} \left. \frac{\partial T(\mathbb{F}_N^t, N)}{\partial t} \right|_{t=0} (\mathbb{F}_N - \mathbb{F}) + \mathcal{E}(\tilde{t}) \\ &= \underbrace{\sqrt{N} \sum_{n=1}^N (\psi_n - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

Inconsistency is suggested if $\mathcal{E}(\tilde{t})$ fails to vanish.

Theorem 3 [Giordano and Broderick, 2023] (sketch):

(Consistency of the von-Mises expansion in finite dimensions)

Under slightly stronger conditions our original finite-dimensional posterior consistency result,

$$\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \rightarrow 0 \quad \text{in the Bayesian von-Mises expansion.}$$

Bayesian von-Mises Expansion Results

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Theorem 4 [Giordano and Broderick, 2023] (sketch):

(Inconsistency of the von-Mises expansion in high dimensional exponential families)

Assume that x_n comes with a equiprobable group assignment $g_n \in 1, \dots, G$.

Conditional on g , x_n is modeled as a finite-dimensional exponential family given λ, γ :

$$\log p(x_n | g_n = g, \gamma, \lambda) = \tau(x_n)^\top \eta_g(\gamma, \lambda) + \text{Constant.}$$

Define the average product of second moments:

$$\mathcal{V}_N(\gamma) := \frac{1}{G} \sum_{g=1}^G \text{tr} \left(\mathbb{E}_{\mathbb{P}(x_n)} [\tau(x_n) \tau(x_n)^\top] \text{Cov}_{p(\lambda|\gamma, \mathbb{F})}(\eta_g(\gamma, \lambda)) \right).$$

If $N \mathbb{E}_{p(\gamma|\mathbb{F})} [\bar{f}(\gamma) \mathcal{V}_N(\gamma)]$ is strictly bounded away from 0 as $N \rightarrow \infty$, then

$$\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \rightarrow \infty \quad \text{in the Bayesian von-Mises expansion.}$$

More experimental results for Gamma–Poisson mixtures

We ran simulations of the Gamma–Poisson mixture with different ratios of N/G (average observations per group).

- When N/G is small:
 - IJ is biased significantly downwards
 - Bootstrap is biased somewhat downwards
- When N/G is larger:
 - Both improve
 - Both remain somewhat biased
 - The IJ and bootstrap perform similarly

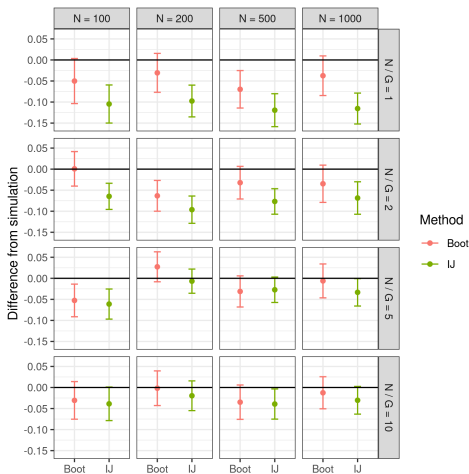


Figure 1: The error of the IJ and bootstrap covariances for different values of N and G . The y-axis shows the difference between $N(V - \hat{V}_{\text{sim}})$, where V is either $\hat{V}_{\text{IJ}}$ or $\hat{V}_{\text{Boot}}$.

Exchangeable units. (A contradiction?)

Negative binomial observations.

Asymptotically linear in w .

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With Gamma REs, one RE per observation, and appropriate prior parameters, these are the same model, with the same $p(\gamma|X)$.

Is $\mathbb{E}_{p(\gamma|X;w)} [\gamma]$ linear in the data weights or not?

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Trick question! We weight a log likelihood contribution, not a datapoint.

$$\log p(X|\gamma; w^m) = \sum_{n=1}^N w_n^m \log p(x_n|\gamma) \quad \log p(X|\gamma, \lambda; w^c) = \sum_{n=1}^N w_n^c \log p(x_n|\lambda, \gamma)$$

The two weightings are not equivalent in general.

What is the right exchangeable unit for a particular problem?

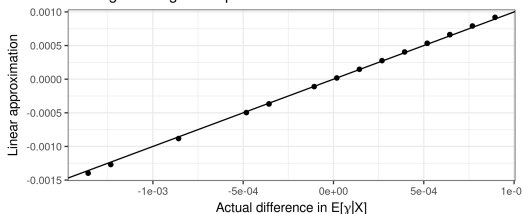
Exchangeable units: Experimental results revisited

Our results were actually computed on **identical datasets** with $G = N$ and $g_n = n$.

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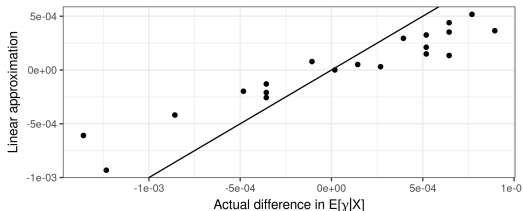
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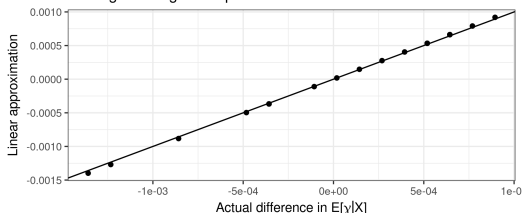
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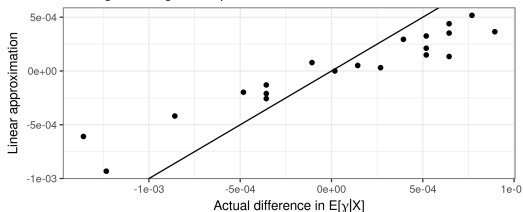
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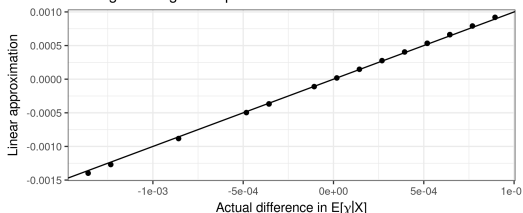
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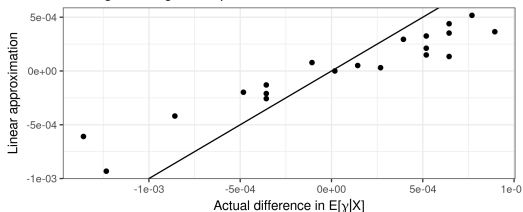
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May still be useful when $p(\lambda|X)$
is *somewhat* concentrated.

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Observations and consequences

- For finite-dimensional models which concentrate asymptotically:
 - Posterior expectations are approximately linear in data weights
 - The linearized variance estimate (infinitesimal jackknife) is consistent
 - The residual of the von Mises expansion vanishes
- For high-dimensional models which marginally concentrate only asymptotically:
 - Posterior expectations are not approximately linear in data weights
 - The linearized variance estimate (infinitesimal jackknife) is inconsistent
 - The residual of the von Mises expansion does not vanish
 - Even if the error $\mathcal{E}(w)$ does not vanish, it can still be small enough in practice.
... Especially given the linear approximation's huge computational advantage.

Observations and consequences

- For finite-dimensional models which concentrate asymptotically:
 - Posterior expectations are approximately linear in data weights
 - The linearized variance estimate (infinitesimal jackknife) is consistent
 - The residual of the von Mises expansion vanishes
- For high-dimensional models which marginally concentrate only asymptotically:
 - Posterior expectations are not approximately linear in data weights
 - The linearized variance estimate (infinitesimal jackknife) is inconsistent
 - The residual of the von Mises expansion does not vanish
 - Even if the error $\mathcal{E}(w)$ does not vanish, it can still be small enough in practice.
... Especially given the linear approximation's huge computational advantage.
- When the weighting is linear, there are many other potential applications:
 - **W-kernel (ongoing work at the IMS!)**
 - **Higher-order confidence intervals (ongoing work at the IMS!)**
 - **Predictive error (ongoing work at the IMS!)**
 - Conformal inference
 - Stochastic coresets
 - Identification of influential subsets
- When the weighting is non-linear, the inconsistency results should apply more widely:
 - The EM algorithm
 - The nonparametric bootstrap
 - Local prior sensitivity measures

Some questions for further discussion

- What (if any) special high structure can give consistency in high dimensions?
- What other tools exist for analyzing the high-dimensional global-local regime?
- Is there any exploitable local information in high-dimensional global-local problems?
- Are there computationally competitive alternatives to differential sensitivity?
- To what extent are these results (negative and positive) relevant for machine learning?

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Preprint: Giordano and Broderick [2023] ([arXiv:2305.06466](#))

R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2023.

B. Kleijn and A. Van der Vaart. The Bernstein-von-Mises theorem under misspecification. *Electronic Journal of Statistics*, 6: 354–381, 2012.